

We first tried a two-channel 1D ResNet, with the following architecture:

- Linearly imputed missing data.
- **Cross-correlation branch** - fixed lag features between channels (± 50 samples), small MLP, 64-d embedding. To explicitly encode inter-channel timing (phase / transit) in a simple, interpretable way and feed it straight into the decision head.
- **Inception block** - multi-scale 1D convs (kernels 1/3/5/7) on 2 into 64 channels.
- **Three ResNet stages** - residual Conv-BN-ReLU blocks (mostly stride 1 early; later stride 2 + dilation for wider receptive field; 64 into 128 into 256 channels).
- **Self-attention** - 4-head attention over the shortened time axis (256-d).
- **Global average pooling** 256-d vector.
- **Output** - 5-threshold ordinal. 5 logits (not 6). Each gets sigmoid $\rightarrow$ probability “yes” for that threshold.

We also tried classical filtering approaches for preprocessing physiological waveforms, creating filtered waveforms, and artificial features.

We also ran KNN, Random Forest and XGBoost on various combinations of these features for baseline comparison.

The Model that worked best was a simple Random Forest Model, using the following preprocessing pipeline and architecture:

- **Data** - Paired aorta + brachial train CSVs (336 timepoints each, 6 age-decade labels) merged per subject.
- **2nd order 8 Hz Chebyshev lowpass filter** (500 Hz sampling) – smooths data without losing features. Taken from standard PPG architecture.
- **Split & imputation:** 80/20 stratified holdout. A **TraceBlockLinearImputer** linearly fills missing values along each trace before the forest sees them.
- **RandomForestClassifier:** With grid search on (3-fold cross validation, macro F1 on the train fold).
 - Final architecture: 100 trees, max depth 20, min_samples_leaf 1, class_weight="balanced".
 - The model was refit on all training subjects before being run on test data.
- **Architecture Summary:** Chebyshev-smoothed dual-site pulse traces, impute, balanced random forest for 6-class age decade, tuned by grid search then refit on full train data.